

1. A. Branch 1: T – T – T – N – N – N (4/6) = 67%  
 Branch 2: T – N – N – T – T – T (4/6) = 67%  
 Branch 3: T – T – N – T – N – T (1/6) = 17%  
 Total 9/18 = 50%  
 B. Branch 1: T – T – T – T – N – N (3/6) = 50%  
 Branch 2: T – T – N – T – T – T (3/6) = 50%  
 Branch 3: T – T – T – T – T – T (3/6) = 50%  
 Total 9/18 = 50%
2. Branch 1 predictions: T – N – T – N – T – N (4/6) = 67%  
 Branch 2 predictions: N – T – T – N – T (3/5) = 60%  
 Branch 3 predictions: N – T – N – T – T – N – T (4/7) = 57%  
 Branch 4 predictions: N – N – N – N (4/4) = 100%  
 Overall accuracy:  
 Total predictions: 22 Total correct: 4 + 3 + 4 + 4 = 15  
 Overall accuracy = Correct/Total = 15/22 = 68%
3. add x15, x12, x11  
 ld x12, 0(x2)  
 nop ; no operation waits for x15  
 ld x13, 4(x15)  
 nop ; no operation  
 nop ; no operation because it waits for x13  
 or x13, x5, x15  
 sd x17, 0(x18)
4. A.  
 ld x19, 0(x8)  
 ld x20, 8(x0)  
 stall  
 addi x21, x19, x20  
 ld x22, 16(x8)  
 stall  
 add x22, x21, x22  
  
 Number of stalls needed = 2  
  
 B.  
 Code Scheduling:  
 ld x19, 0(x8)  
 ld x20, 8(x0)  
 ld x22, 16(x8)  
 addi x21, x19, x20  
 add x22, x21, x22
5. A.  
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| EXECUTION TABLE  |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|------------------|------------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
| FULL LOOPS       | CPU Cycles |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
| Instruction      | 1          | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| addi sp, sp, -24 | F          | D | X | M | W |   |   |   |   |    |    |    |    |    |    |    |
| sd t0, 16(sp)    |            | F | D | X | M | W |   |   |   |    |    |    |    |    |    |    |
| sd t1, 8(sp)     |            |   | F | D | X | M | W |   |   |    |    |    |    |    |    |    |
| sd s4, 0(sp)     |            |   |   | F | D | X | M | W |   |    |    |    |    |    |    |    |
| add t0, a0, a1   |            |   |   |   | F | D | X | M | W |    |    |    |    |    |    |    |
| add t1, a2, a3   |            |   |   |   |   | F | D | X | M | W  |    |    |    |    |    |    |
| sub s4, t0, t1   |            |   |   |   |   |   | F | D | X | M  | W  |    |    |    |    |    |
| addi a0, s4, 0   |            |   |   |   |   |   |   | F | D | X  | M  | W  |    |    |    |    |
| ld s4, 0(sp)     |            |   |   |   |   |   |   |   | F | D  | X  | M  | W  |    |    |    |
| ld t1, 8(sp)     |            |   |   |   |   |   |   |   |   | F  | D  | X  | M  | W  |    |    |
| ld t0, 16(sp)    |            |   |   |   |   |   |   |   |   |    | F  | D  | X  | M  | W  |    |
| addi sp, sp, 24  |            |   |   |   |   |   |   |   |   |    |    | F  | D  | X  | M  | W  |

The elimination of the 3 load-use stalls is achieved by directly forwarding the loaded values to the dependent instructions, bypassing the need to wait for the load instruction to complete its writeback stage. As a result, back-to-back execution can occur seamlessly without any interruptions.

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| EXECUTION TABLE  |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
|------------------|------------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|
| FULL LOOPS       | CPU Cycles |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| Instruction      | 1          | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
| addi sp, sp, -24 | F          | D | X | M | W |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| sd t0, 16(sp)    |            | F | - | - | D | X | M | W |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| sd t1, 8(sp)     |            |   |   |   | F | D | X | M | W |    |    |    |    |    |    |    |    |    |    |    |    |    |
| sd s4, 0(sp)     |            |   |   |   |   | F | D | X | M | W  |    |    |    |    |    |    |    |    |    |    |    |    |
| add t0, a0, a1   |            |   |   |   |   |   | F | D | X | M  | W  |    |    |    |    |    |    |    |    |    |    |    |
| add t1, a2, a3   |            |   |   |   |   |   |   | F | D | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |
| sub s4, t0, t1   |            |   |   |   |   |   |   |   | F | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |
| addi a0, s4, 0   |            |   |   |   |   |   |   |   |   |    |    | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |
| ld s4, 0(sp)     |            |   |   |   |   |   |   |   |   |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |
| ld t1, 8(sp)     |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |
| ld t0, 16(sp)    |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |
| addi sp, sp, 24  |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |

In all three scenarios involving lw-sd, lw-sub, and lw-addi, there is a need for 3 cycles to resolve load-use stalls. These instructions depend on values loaded by a previous instruction, and since the loaded values are not forwarded, they are required to wait for the load instruction to finish before they can proceed with their execution.

B.  
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| Instruction       | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|-------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|
| addi ra, x0, 1    | F | D | X | M | W |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| add sp, ra, ra    |   | F | D | X | M | W |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| sub gp, sp, ra    |   |   | F | D | X | M | W |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| slt tp, ra, sp    |   |   |   | F | D | X | M | W |   |    |    |    |    |    |    |    |    |    |    |    |    |    |
| addi t0, sp, -3   |   |   |   |   | F | D | X | M | W |    |    |    |    |    |    |    |    |    |    |    |    |    |
| andi t0, t0, 2032 |   |   |   |   |   | F | D | X | M | W  |    |    |    |    |    |    |    |    |    |    |    |    |
| ori t0, t0, 15    |   |   |   |   |   |   | F | D | X | M  | W  |    |    |    |    |    |    |    |    |    |    |    |
| add t0, t0, t0    |   |   |   |   |   |   |   | F | D | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |
| addi t1, x0, 1024 |   |   |   |   |   |   |   |   | F | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |
| lw t2, 0(t1)      |   |   |   |   |   |   |   |   |   | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |
| add s0, t2, sp    |   |   |   |   |   |   |   |   |   |    | F  | -  | D  | X  | M  | W  |    |    |    |    |    |    |
| sw s0, 4(t1)      |   |   |   |   |   |   |   |   |   |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |
| slt s1, sp, ra    |   |   |   |   |   |   |   |   |   |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |
| or a0, t0, s0     |   |   |   |   |   |   |   |   |   |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |
| xor a1, t1, sp    |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |
| ori a2, x0, 0     |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |
| ori a3, x0, 2     |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |

With forwarding, 7 stalls are mitigated by resolving the load-use hazards. However, the load-use hazard involving the load (lw) instruction and the subsequent add (add) instruction still persists because forwarding cannot eliminate stalls across the store buffer. Consequently, only the lw-add stall remains, and the other 7 stalls are successfully averted.

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| EXECUTION TABLE   |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
|-------------------|------------|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--|
| FULL LOOPS        | CPU Cycles |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| Instruction       | 1          | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 |  |
| addi ra, x0, 1    | F          | D | X | M | W |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| add sp, ra, ra    |            | F | - | - | D | X | M | W |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| sub gp, sp, ra    |            |   |   |   | F | - | - | D | X | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| slt tp, ra, sp    |            |   |   |   |   |   | F | D | X | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| addi t0, sp, -3   |            |   |   |   |   |   |   | F | D | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| andi t0, t0, 2032 |            |   |   |   |   |   |   |   | F | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| ori t0, t0, 15    |            |   |   |   |   |   |   |   |   | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| add t0, t0, t0    |            |   |   |   |   |   |   |   |   |    | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| addi t1, x0, 1024 |            |   |   |   |   |   |   |   |   |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| lw t2, 0(t1)      |            |   |   |   |   |   |   |   |   |    |    |    | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| add s0, t2, sp    |            |   |   |   |   |   |   |   |   |    |    |    |    | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| sw s0, 4(t1)      |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    | F  | -  | -  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| slt s1, sp, ra    |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| or a0, t0, s0     |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |    |  |
| xor a1, t1, sp    |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |    |  |
| ori a2, x0, 0     |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |    |  |
| ori a3, x0, 2     |            |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    | F  | D  | X  | M  | W  |    |    |    |    |    |    |    |    |    |    |  |

There are 8 stall cycles required. 6 for the load-use cases of lw-add, lw-sub, lw-andi, lw-ori, lw-add and 1 for the lw-sw case.